Assignment Chapter 5: Solution

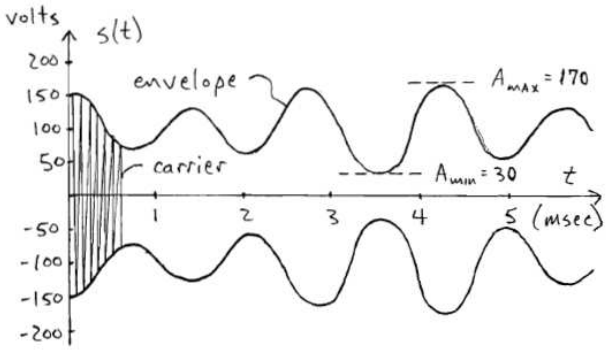
Problem 5-2:
Part (a):

- The complex envelope is given by:

$$g(t) = A_c[1 + m(t)] = 100\,[1 + 0.2\sin(\omega_1 t) + 0.5\cos(\omega_2 t)]$$

  and the AM signal is given by: $s(t) = g(t)\cos(\omega_c t)$.

- These signals are plotted in the figure below (using Matlab):



  - Note that this part can not be solved analytically. A programmable calculator or Matlab is needed.

Part (b):

- From the figure, we observe that the maximum and minimum values of $g(t)$ are given by $A_{max} = 170$ and $A_{min} = 30$ implying that the modulation percentage is given by:

$$\begin{aligned}\%\text{ modulation} &= \frac{A_{max} - A_{min}}{2A_c} \times 100 \\ &= \frac{170 - 30}{200} \times 100 = 70\%\end{aligned}$$

Part (c):

- The spectrum of $g(t)$ is given by:

$$\begin{aligned}G(f) &= 100\,[\delta(f) + 0.1j\delta(f + f_1) - 0.1j\delta(f - f_1) + 0.25\delta(f + f_2) + 0.25\delta(f - f_2)] \\ &= 100\delta(f) + 10j\delta(f + f_1) - 10j\delta(f - f_1) + 25\delta(f + f_2) + 25\delta(f - f_2)\end{aligned}$$

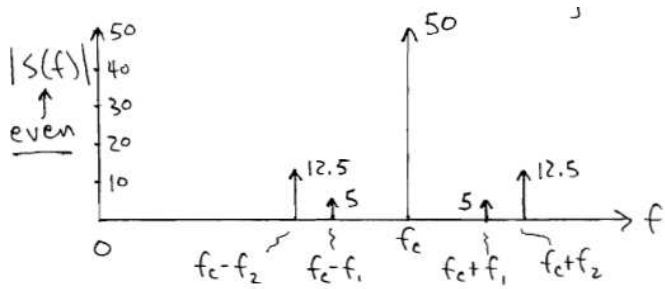
  implying that:

$$\begin{aligned}S(f) &= \frac{1}{2}[G(f - f_c) + G[f + f_c)] \\ &= 50\delta(f - f_c) + 5j\delta(f - (f_c - f_1)) - 5j\delta(f - (f_c + f_1)) + 12.5\delta(f - (f_c - f_2)) + 12.5\delta(f - (f_c + f_2)) \\ &+ 50\delta(f + f_c) + 5j\delta(f + f_c + f_1) - 5j\delta(f + f_c - f_1) + 12.5\delta(f + f_c + f_2) + 12.5\delta(f + f_c - f_2)\end{aligned}$$

- The magnitude spectrum located at positive frequencies is given by:

  $$|S(f)| = 50\delta(f-f_c)+5\delta(f-(f_c-f_1))+5\delta(f-(f_c+f_1))+12.5\delta(f-(f_c-f_2))+12.5\delta(f-(f_c+f_2))$$

  This spectrum is shown in the figure below:



Problem 5-3:
Part (a):

- The average power of the AM signal is given by:

$$\begin{aligned}\left\langle s^2(t)\right\rangle &= \frac{1}{2}\left\langle g^2(t)\right\rangle = \frac{1}{2}A_c^2\left\langle [1+m(t)]^2\right\rangle \\ &= \frac{1}{2}A_c^2 + A_c^2\left\langle m(t)\right\rangle + \frac{1}{2}A_c^2\left\langle m^2(t)\right\rangle\end{aligned}$$

where:

$$\begin{aligned}\langle m(t)\rangle &= 0.2\underbrace{\langle\sin(\omega_1 t)\rangle}_{=0} + 0.5\underbrace{\langle\cos(\omega_2 t)\rangle}_{=0} = 0 \\ \left\langle m^2(t)\right\rangle &= 0.04\left\langle\sin^2(\omega_1 t)\right\rangle + 0.25\left\langle\cos^2(\omega_2 t)\right\rangle + 0.2\left\langle\sin(\omega_1 t)\cos(\omega_2 t)\right\rangle \\ &= 0.04\underbrace{\left\langle\frac{1}{2}-\frac{1}{2}\cos(2\omega_1 t)\right\rangle}_{=\frac{1}{2}} + 0.25\underbrace{\left\langle\frac{1}{2}+\frac{1}{2}\cos(2\omega_2 t)\right\rangle}_{=\frac{1}{2}} \\ &\quad + 0.2\underbrace{\left\langle\frac{1}{2}\sin((\omega_1+\omega_2)t)+\frac{1}{2}\sin((\omega_1-\omega_2)t)\right\rangle}_{=0} \\ &= 0.02 + 0.125 = 0.145\end{aligned}$$

- Consequently:

$$\left\langle s^2(t)\right\rangle = \frac{1}{2}100^2 + \frac{1}{2}100^2(0.145) = 5725 \ \text{ W}$$

- The average power fed to the load of 50 Ω is given by:

$$P_{av} = \frac{\left\langle s^2(t)\right\rangle}{50} = \frac{5725}{50} = 114.5 \ \text{ W}$$

Part (b):

- The normalized PEP of the AM signal is given by:

$$P_{PEP,n} = \frac{1}{2}\left\{\max|g(t)|\right\}^2 = \frac{1}{2}(170)^2 = 14450 \ \text{ W}$$

implying that the PEP is equal to:

$$P_{PEP} = \frac{P_{PEP,n}}{50} = \frac{14450}{50} = 289 \ \text{ W}$$

Problem 5-5:
Part (a):

- The power of 50 kW corresponds to the power of the carrier. For a charge of 50 Ω:
$$P_c = \frac{1}{50}\left[\frac{1}{2}A_c^2\right] = 50,000 \quad \Rightarrow \quad A_c = 2236 \ \text{ V}$$

- Consequently:
$$g(t) = A_c[1 + m(t)] = 2236\,[1 + A_1\cos(\omega_1 t) + A_1\cos(2\omega_1 t)]$$

Part (b):

- The maximum value of $m(t)$ is: $\max[m(t)] = 2A_1$ (which can be obtained, for example, by setting $t = 0$).

- To determine the minimum value of $m(t)$, we set $m'(t)$ equal to zero:
$$\begin{aligned} m'(t) &= -A_1\omega_1\sin(\omega_1 t) - 2A_1\omega_1\sin(2\omega_1 t) = 0 \\ \Rightarrow \quad \frac{\sin(2\omega_1 t)}{\sin(\omega_1 t)} &= -\frac{1}{2} \quad \Rightarrow \quad \frac{2\sin(\omega_1 t)\cos(\omega_1 t)}{\sin(\omega_1 t)} = -\frac{1}{2} \\ \Rightarrow \quad \cos(\omega_1 t) &= -\frac{1}{4} \quad \Rightarrow \quad \omega_1 t = 1.8235 \ \text{ rad} \end{aligned}$$
implying that the minimum value of $m(t)$ is given by:
$$\min[m(t)] = A_1\,[\cos(1.8235) + \cos(2 \times 1.8235)] = -1.125A_1$$

- Consequently, for the modulation percentage to be equal to 90%:
$$\begin{aligned} \%\ \text{modulation} &= \frac{\max[m(t)] - \min[m(t)]}{2} \times 100 = 90\% \\ \Rightarrow \quad \frac{2A_1 - (-1.125)A_1}{2} \times 100 = 90\% \quad &\Rightarrow \quad A_1 = 0.576 \ \text{ V} \end{aligned}$$

Part (c):

- The voltage fed into the load is given by:
$$s(t) = 2236[1 + m(t)]\cos(\omega_c t)$$

- The peak voltage fed into the load is:
$$A_{max} = 2236[1 + \max[m(t)]] = 2236[1 + 2 \times 0.576] = 4811.9 \ \text{ V}$$
implying that the peak current is:
$$I_{max} = \frac{A_{max}}{50} = 96.238 \ \text{ A}$$

- The mean voltage fed into the load is:

$$
\begin{aligned}
A_{av} &= \langle 2236[1+m(t)]\cos(\omega_c t)\rangle \\
&= 2236\,\langle [1+0.576\cos(\omega_1 t)+0.576\cos(2\omega_1 t)]\cos(\omega_c t)\rangle \\
&= 2236\,\underbrace{\langle \cos(\omega_c t)\rangle}_{=0} + 1288\,\underbrace{\langle \cos(\omega_1 t)\cos(\omega_c t)\rangle}_{=\left\langle \frac{1}{2}\cos((\omega_c+\omega_1)t)+\frac{1}{2}\cos((\omega_c-\omega_1)t)\right\rangle=0} \\
&\quad + 1288\,\underbrace{\langle \cos(2\omega_1 t)\cos(\omega_c t)\rangle}_{=\left\langle \frac{1}{2}\cos((\omega_c+2\omega_1)t)+\frac{1}{2}\cos((\omega_c-2\omega_1)t)\right\rangle=0}
\end{aligned}
$$

  - The last equalities follow since $\omega_c \gg \omega_1$ implying that $\omega_c - \omega_1 \neq 0$ and $\omega_c - 2\omega_1 \neq 0$.

  Consequently, the mean current is:

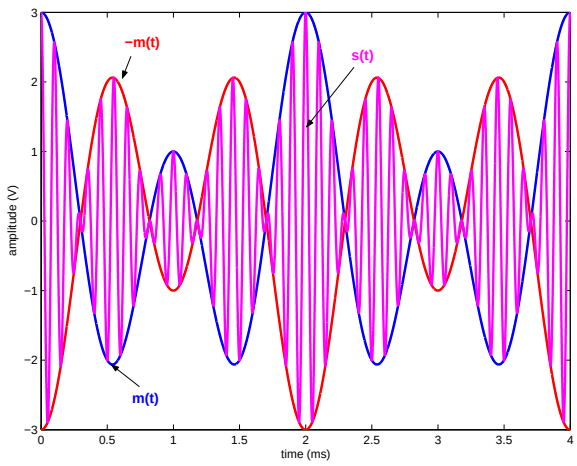
$$I_{av} = 0 \quad \text{A}$$

Problem 5-7:
Part (a):

- The DSB-SC signal is given by:

$$s(t) = A_c m(t)\cos(\omega_c t) = [\cos(1000\pi t) + 2\cos(2000\pi t)]\cos(\omega_c t)$$

- This signal is plotted in the figure below (using Matlab):



Part (b):

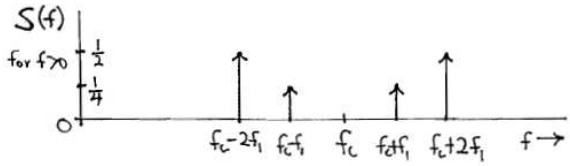
- The Fourier transform of $s(t)$ is given by:

$$
\begin{aligned}
S(f) &= \left[\frac{1}{2}\delta(f-f_1)+\frac{1}{2}\delta(f+f_1)+\delta(f-2f_1)+\delta(f+2f_1)\right] * \left[\frac{1}{2}\delta(f-f_c)+\frac{1}{2}\delta(f+f_c)\right] \\
&= \frac{1}{4}\delta(f-(f_c+f_1))+\frac{1}{4}\delta(f-(f_c-f_1))+\frac{1}{2}\delta(f-(f_c+2f_1))+\frac{1}{2}\delta(f-(f_c-2f_1)) \\
&\quad + \frac{1}{4}\delta(f+f_c-f_1)+\frac{1}{4}\delta(f+f_c+f_1)+\frac{1}{2}\delta(f+f_c-2f_1)+\frac{1}{2}\delta(f+f_c+2f_1)
\end{aligned}
$$

- The magnitude spectrum located at positive frequencies is given by:

$$|S(f)| = \frac{1}{4}\delta(f-(f_c+f_1))+\frac{1}{4}\delta(f-(f_c-f_1))+\frac{1}{2}\delta(f-(f_c+2f_1))+\frac{1}{2}\delta(f-(f_c-2f_1))$$

  This spectrum is plotted in the figure below:



Part (c):

- The average power is given by:

$$\begin{aligned}\langle s^2(t)\rangle &= \frac{1}{2}\langle g^2(t)\rangle = \frac{1}{2}A_c^2\langle m^2(t)\rangle \\ &= \frac{1}{2}\underbrace{\langle\cos^2(\omega_1 t)\rangle}_{=\frac{1}{2}} + 2\underbrace{\langle\cos^2(2\omega_1 t)\rangle}_{=\frac{1}{2}} + 2\underbrace{\langle\cos(\omega_1 t)\cos(2\omega_1 t)\rangle}_{=0} \\ &= \frac{1}{4}+1 = 1.25 \ \text{ W}\end{aligned}$$

Part (d):

- The maximum value of $g(t)$ is $A_{max} = 3$ implying that:

$$PEP = \frac{A_{max}^2}{2} = 4.5 \ \text{ W}$$

Problem 5-8:

- Assume that:

$$\max[m(t)] = 1 \quad ; \quad \langle m(t)\rangle = 0$$

- For a DSB-SC signal: $s(t) = A_c m(t)\cos(\omega_c t)$ implying that the peak value is:

$$A_p = A_c \max[m(t)] = A_c$$

  The sideband power of this signal (which is the same as the power of the signal) is given by:

$$P_{DSB,\text{sideband}} = \frac{1}{2}\langle g^2(t)\rangle = \frac{1}{2}A_p^2\langle m^2(t)\rangle$$

- For an AM signal: $s(t) = A_c[1+m(t)]\cos(\omega_c t)$ implying that the peak value is:

$$A_p = A_c[1+\max[m(t)]] = 2A_c \quad \Rightarrow \quad A_c = \frac{A_p}{2}$$

The power of this signal is given by:

$$P_{AM} = \frac{1}{2}\left\langle g^2(t)\right\rangle = \underbrace{\frac{1}{2}A_c^2}_{\text{carrier power}} + A_c^2 \underbrace{\left\langle m(t)\right\rangle}_{=0} + \underbrace{\frac{1}{2}A_c^2\left\langle m^2(t)\right\rangle}_{\text{sideband power}}$$

implying that the sideband power is:

$$P_{AM,\text{sideband}} = \frac{1}{2}A_c^2\left\langle m^2(t)\right\rangle = \frac{1}{8}A_p^2\left\langle m^2(t)\right\rangle$$

- Consequently:

$$\frac{P_{DSB,\text{sideband}}}{P_{AM,\text{sideband}}} = \frac{\frac{1}{2}A_p^2\left\langle m^2(t)\right\rangle}{\frac{1}{8}A_p^2\left\langle m^2(t)\right\rangle} = 4$$

Problem 5-13:
Part (a):

- Applying a $-\pi/2$ phase shift to $m(t)$ results in:

$$\hat{m}(t) = 5\cos(\omega_1 t - \pi/2) = 5\sin(\omega_1 t)$$

Part (b):

- The LSSB signal is given by:

$$\begin{aligned} s(t) &= A_c\left[m(t)\cos(\omega_c t) + \hat{m}(t)\sin(\omega_c t)\right] \\ &= 5\cos(\omega_1 t)\cos(\omega_c t) + 5\sin(\omega_1 t)\sin(\omega_c t) \end{aligned}$$

Note that $s(t)$ can be written as:

$$s(t) = 5\cos[(\omega_c - \omega_1)t]$$

Part (c):

- Given that the RMS value of a sinusoidal signal is $1/\sqrt{2}$, then the RMS value of $s(t)$ is given by:

$$V_{RMS} = \frac{5}{\sqrt{2}} \text{ V}$$

Part (d):

- The peak value of $s(t)$ is $V_p = 5$ V.

Part (e):

- The normalized average power is given by:

$$P_{av} = \left\langle s^2(t)\right\rangle = 25\underbrace{\left\langle\cos^2[(\omega_c - \omega_1)t]\right\rangle}_{=\frac{1}{2}} = 12.5 \text{ W}$$

Second Method:

- The complex envelope is given by: $g(t) = A_c\,[m(t) + j\hat{m}(t)]$. Since both $m(t)$ and $\hat{m}(t)$ are real:

$$|g(t)|^2 = A_c^2[m^2(t) + \hat{m}^2(t)] = 25\cos^2(\omega_1 t) + 25\sin^2(\omega_1 t) = 25$$

implying that:

$$\left\langle s^2(t)\right\rangle = \frac{1}{2}\left\langle |g(t)|^2\right\rangle = \frac{25}{2} = 12.5 \ \ \text{W}$$

Part (f):

- Since $|g(t)| = 5$, then $A_{max} = \max[|g(t)| = 5$ implying that:

$$PEP = \frac{A_{max}^2}{2} = \frac{25}{2} = 12.5 \ \ \text{W}$$

Problem 5-14:
Part (a):

- $\hat{m}(t)$ is given by:

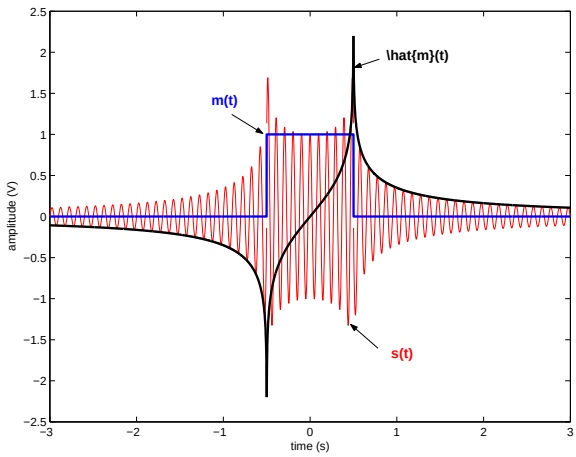
$$\begin{aligned}
\hat{m}(t) &= m(t) * \frac{1}{\pi t} \\
&= \int_{-\infty}^{+\infty} m(\lambda)\frac{1}{\pi(t-\lambda)}\mathrm{d}\lambda = \int_{-T/2}^{+T/2} \frac{1}{\pi(t-\lambda)}\mathrm{d}\lambda \\
&= \int_{t+T/2}^{t-T/2} \frac{-1}{\pi\lambda_1}\mathrm{d}\lambda_1 = -\frac{1}{\pi}\ln\left|\frac{t-T/2}{t+T/2}\right| \\
&= \frac{1}{\pi}\ln\left|\frac{t+T/2}{t-T/2}\right| = \frac{1}{\pi}\ln\left|\frac{2t+T}{2t-T}\right|
\end{aligned}$$

Part (b):

- Consider a USSB-AM signal. This signal is given by:

$$\begin{aligned}
s(t) &= A_c\left[m(t)\cos(\omega_c t) - \hat{m}(t)\sin(\omega_c t)\right] \\
&= \mathrm{rect}\left(\frac{t}{T}\right)\cos(\omega_c t) - \frac{1}{\pi}\ln\left|\frac{2t+T}{2t-T}\right|\sin(\omega_c t)
\end{aligned}$$

This signal is plotted below for $T = 1$ (using Matlab):



Part (c):

- From the above figure, we observe that the peak (maximum) value of $s(t)$ is $+\infty$.

Problem 5-15:
Part (a):

- The spectrum of the modulating signal is given by:

$$m(t) = \text{rect}\,(t/T) \quad \Rightarrow \quad M(f) = T\text{sinc}(Tf)$$

- For a USSB-AM signal, the spectrum of the complex envelope is given by:

$$G(f) = \begin{cases} 2A_c M(f) = 2A_c T\text{sinc}(Tf), & f > 0; \\ 0, & f < 0. \end{cases}$$

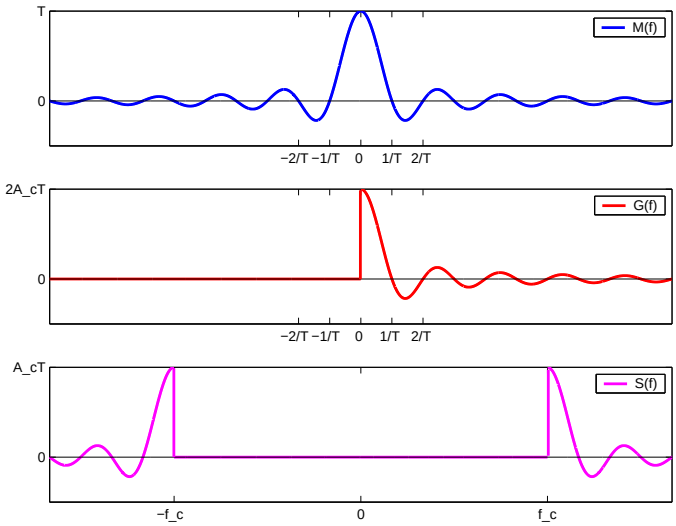
- Consequently:

$$\begin{aligned} S(f) &= \frac{1}{2}\left[G(f - f_c) + G^*(-f - f_c)\right] \\ &= \begin{cases} A_c T\text{sinc}[T(f - f_c)], & f > f_c; \\ [A_c T\text{sinc}[T(-f - f_c)]]^* = A_c T\text{sinc}[T(f + f_c)], & f < -f_c; \\ 0, & \text{otherwise}; \end{cases} \end{aligned}$$

  - Note that the sinc function is even.

Part (b):

- $M(f)$, $G(f)$ and $S(f)$ are plotted in the figurer below:



Problem 5-16:
Part (a):

- The Fourier transform of $m(t)$ is given by:

$$m(t) = \text{sinc}(at) \quad \Rightarrow \quad M(f) = \frac{1}{a}\text{rect}\left(\frac{f}{a}\right)$$

$M(f)$ can be separated into two parts $M^+(f)$ and $M^-(f)$ located at positive and negative frequencies respectively:

$$M(f) = \underbrace{\frac{1}{a}\text{rect}\left(\frac{f - a/4}{a/2}\right)}_{M^+(f)} + \underbrace{\frac{1}{a}\text{rect}\left(\frac{f + a/4}{a/2}\right)}_{M^-(f)}$$

- Consequently:

$$\begin{aligned}\hat{M}(f) = M(f)H(f) &= M(f)\begin{cases} -j, & f > 0; \\ j, & f < 0. \end{cases} \\ &= -jM^+(f) + jM^-(f)\end{aligned}$$

resulting in:

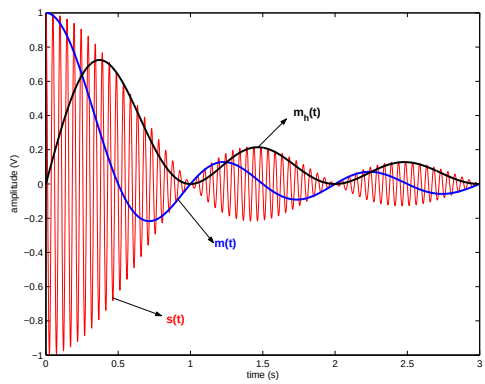
$$\begin{aligned}\hat{m}(t) &= -j\frac{1}{a}\frac{a}{2}\text{sinc}\left(\frac{a}{2}t\right)e^{j2\pi\frac{a}{4}t} + j\frac{1}{a}\frac{a}{2}\text{sinc}\left(\frac{a}{2}t\right)e^{-j2\pi\frac{a}{4}t} \\ &= \text{sinc}\left(\frac{a}{2}t\right)\left[\frac{e^{j\pi\frac{a}{2}t} - e^{-j\pi\frac{a}{2}t}}{2j}\right] = \text{sinc}\left(\frac{a}{2}t\right)\sin\left(\pi\frac{a}{2}t\right) = \frac{\sin^2\left(\pi\frac{a}{2}t\right)}{\pi\frac{a}{2}t}\end{aligned}$$

Part (b):

- The USSB-AM signal is given by:

$$\begin{aligned}s(t) &= A_c\left[m(t)\cos(\omega_c t) - \hat{m}(t)\sin(\omega_c t)\right] \\ &= \text{sinc}(at)\cos(\omega_c t) - \frac{\sin^2\left(\pi\frac{a}{2}t\right)}{\pi\frac{a}{2}t}\sin(\omega_c t)\end{aligned}$$

- This signal is plotted in the figure below for $a = 2$ and $f_c = 20$ Hz (using Matlab):



Problem 5-18:
Parts (a), (b), (c):

- The input signal is given by:

$$v_A(t) = m(t)\cos(\omega_{if}t) \underbrace{\overbrace{\pm}^{LSSB}}_{USSB} \hat{m}(t)\sin(\omega_{if}t)$$

  - The + sign corresponds to a LSSB signal.
  - The - sign corresponds to a USSB signal.

- The signal at the output of the local oscillator is given by:

$$\begin{aligned} & v_D(t) = \cos(\omega_{if}t) \quad ; \quad \omega_{if} = 2\pi(455) \ \text{ kHz} \\ \Rightarrow \ & v_E(t) = \cos(\omega_{if}t - \pi/2) = \sin(\omega_{if}t) \end{aligned}$$

- The signals at points B and C are given by:

$$\begin{aligned} v_B(t) &= v_A(t)v_D(t) \\ &= m(t)\cos^2(\omega_{if}t) \pm \hat{m}(t)\sin(\omega_{if}t)\cos(\omega_{if}t) \\ &= \frac{1}{2}m(t) + \underbrace{\frac{1}{2}m(t)\cos(2\omega_{if}t)}_{\text{removed by the LPF}} \pm \underbrace{\frac{1}{2}\hat{m}(t)\sin(2\omega_{if}t)}_{\text{removed by the LPF}} \\ \Rightarrow \quad v_C(t) &= \frac{1}{2}m(t) \end{aligned}$$

- The signals at points F and G are given by:

$$\begin{aligned} v_F(t) &= v_A(t)v_E(t) \\ &= m(t)\cos(\omega_{if}t)\sin(\omega_{if}t) \pm \hat{m}(t)\sin^2(\omega_{if}t) \\ &= \underbrace{\frac{1}{2}m(t)\sin(2\omega_{if}t)}_{\text{removed by the LPF}} \pm \frac{1}{2}\hat{m}(t) \mp \underbrace{\frac{1}{2}\hat{m}(t)\cos(2\omega_{if}t)}_{\text{removed by the LPF}} \\ \Rightarrow \quad v_G(t) &= \pm\frac{1}{2}\hat{m}(t) \end{aligned}$$

- We will next determine the value of $v_H(t)$. $\hat{m}(t)$ is obtained by applying a $-\pi/2$ phase shift to $m(t)$. Consequently, applying a $-\pi/2$ phase shift to $\hat{m}(t)$ is equivalent to an overall phase shift of $-\pi$ applied to $m(t)$ (thus resulting in $-m(t)$). Therefore:

$$v_H(t) = \mp\frac{1}{2}m(t)$$

- Finally:

$$\begin{aligned} v_I(t) &= v_C(t) - v_H(t) \\ &= \frac{1}{2}m(t) \pm \frac{1}{2}m(t) = \begin{cases} m(t), & \text{LSSB;} \\ 0, & \text{USSB.} \end{cases} \end{aligned}$$

  Consequently, the above detector can be used to detect LSSB signals.

- To receive USSB signals, the subtractor must be replaced by an adder resulting in:

$$\begin{aligned} v_I(t) &= v_C(t) + v_H(t) \\ &= \frac{1}{2}m(t) \mp \frac{1}{2}m(t) = \begin{cases} 0, & \text{LSSB;} \\ m(t), & \text{USSB.} \end{cases} \end{aligned}$$

Part (d):

- The filters must have a bandwidth of 3 kHz. They should be centered at $f_{if} - 1.5$ kHz for LSSB and $f_{if} + 1.5$ kHz for USSB.

Problem 5-20:
Part (a):

- The amplitude of $s(t)$ is not modulated implying that the percentage of AM modulation is 0%.

Part (b):

- The normalized average power is given by:

$$P_{av} = \left\langle s^2(t) \right\rangle = 100 \underbrace{\left\langle \cos^2[2\pi 10^8 t + 10\cos(2\pi 10^3 t)] \right\rangle}_{=\frac{1}{2}} = 50 \ \ \text{W}$$

Part (c):

- The maximum phase deviation is given by:

$$\Delta\theta_{max} = \max\left[10\cos(2\pi 10^3 t)\right] = 10 \ \ \text{rad}$$

Part (d):

- The instantaneous frequency of $s(t)$ is given by:

$$\begin{aligned} f_i(t) &= 10^8 + \frac{1}{2\pi}\frac{\mathrm{d}\left[10\cos(2\pi 10^3 t)\right]}{\mathrm{d}t} \\ &= 10^8 - \frac{1}{2\pi}10(2\pi 10^3)\sin(2\pi 10^3 t) \\ &= 10^8 - 10^4 \sin(2\pi 10^3 t) \end{aligned}$$

  implying that the frequency deviation is given by:

$$f_d(t) = -10^4 \sin(2\pi 10^3 t)$$

- Consequently, the maximum frequency deviation is given by:

$$\Delta F_{max} = \max\left[f_d(t)\right] = 10^4 \ \ \text{Hz} = 10 \ \ \text{kHz}$$

Problem 5-22:
Part (a):

- The modulating signal is given by $m(t) = 4\sin(2\pi 10^3 t)$. The peak frequency deviation is given by:

$$\Delta F = \underbrace{\frac{1}{2\pi} \underbrace{D_f}_{\text{rad/(V.s)}\equiv\text{rad.Hz/V}}}_{\text{Hz/V}} \underbrace{\max[m(t)]}_{=4 \ \text{V}}$$

  implying that:

$$\Delta F = [50 \ \text{Hz/V}] \times [4 \ \text{V}] = 200 \ \text{Hz}$$

Part (b):

- The modulation index is given by:

$$\beta_f = \frac{\Delta F}{B} = \frac{200}{1000} = 0.2$$

Problem 5-23:
Part (a):

- Using Carson's rule:

$$B_T = 2(\beta + 1)B = 2(\beta + 1)f_m = 2(2 + 1)15 = 90 \ \text{kHz}$$

Part (b):

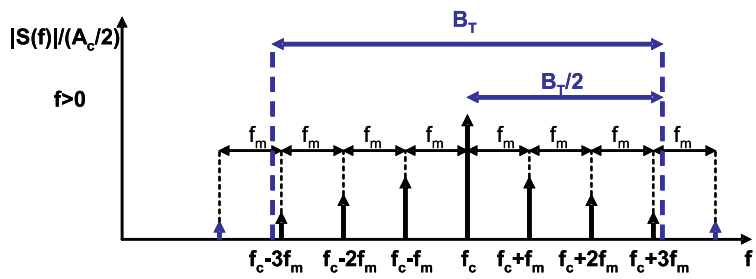
- The complex envelope of PM and FM signals is given by: $g(t) = A_c e^{j\theta(t)}$ implying that the total average power is given by:

$$P_{av} = \frac{1}{2}\left\langle |g(t)|^2 \right\rangle = \frac{1}{2}\left\langle A_c^2 \right\rangle = \frac{1}{2}A_c^2$$

- Given that two consecutive frequency components are separated by f_m, then the number of components above (or below) the carrier frequency is given by:

$$N = \frac{\frac{B_T}{2}}{f_m} = \frac{45}{15} = 3$$

This is better illustrated in the figure below:



- Given that the magnitude spectrum of the modulated signal <u>for positive frequencies</u> is given by:

$$|S(f)| = \frac{1}{2}A_c\left[|J_0(\beta)|\delta\left(f-f_c\right)+\sum_{n=1}^{+\infty}|J_n(\beta)|\left[\delta\left(f-(f_c+nf_m)\right)+\delta\left(f-(f_c-nf_m)\right)\right]\right]$$

then the power contained in the bandwidth B_T is:

$$P_T = 2\times\left(\frac{1}{2}A_c\right)^2\left[|J_0(\beta)|^2+2\sum_{n=1}^{N=3}|J_n(\beta)|^2\right]$$

  - The multiplying factor of 2 was added in order to take the negative frequency components into consideration.

Consequently:

$$\begin{aligned}P_T &= \frac{1}{2}A_c^2\left[J_0^2(2)+2[J_1^2(2)+J_2^2(2)+J_3^2(2)]\right]\\ &= \frac{1}{2}A_c^2\left[(0.224)^2+2[(0.577)^2+(0.353)^2+(0.129)^2]\right]\\ &= \frac{1}{2}A_c^2\left[0.9985\right]\end{aligned}$$

implying that the percentage of the power that lies within the Carson's bandwidth is:

$$\%\text{ Power} = \frac{P_T}{P_{av}}\times 100 = \frac{\frac{1}{2}A_c^2\left[0.9985\right]}{\frac{1}{2}A_c^2}\times 100 = 99.85\%$$

Problem 5-26:
Part (a):

- The phase deviation constant is $D_p = 100$ rad/V. Consequently:

$$\begin{aligned}&\theta(t) = D_p m(t) = 20\cos(\omega_1 t)\\ \Rightarrow\ &m(t) = 0.2\cos(\omega_1 t)\end{aligned}$$

- The peak value of $m(t)$ is 0.2 V and its frequency is $f_1 = 1$ kHz.

Part (b):

- The frequency deviation constant is $D_f = 10^6$ rad/(V.s). Consequently:

$$\theta(t) = D_f \int_{-\infty}^{t} m(\tau)\mathrm{d}\tau = 20\cos(\omega_1 t)$$
$$\Rightarrow \quad m(t) = \frac{20}{D_f}\frac{\mathrm{d}[\cos(\omega_1 t)]}{\mathrm{d}t} = -\frac{20\omega_1}{D_f}\sin(\omega_1 t)$$
$$= -\frac{20(2\pi 1000)}{10^6}\sin(\omega_1 t) = -0.1257\sin(\omega_1 t)$$

- The peak value of $m(t)$ is 0.1257 V and its frequency is $f_1 = 1$ kHz.

Part (c):

- For both the PM and the FM signals, the average power is given by:

$$P_{av} = \frac{\frac{1}{2}A_c^2}{50} = \frac{\frac{1}{2}(500)^2}{50} = 2500 \ \ \mathrm{W}$$

- In the same way, the PEP is given by:

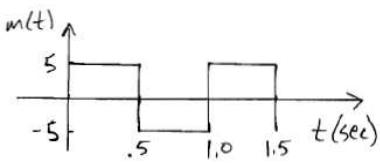
$$PEP = \frac{\frac{1}{2}\max[|g(t)|^2]}{50}$$

since $g(t) = A_c e^{j\theta(t)}$, then $|g(t)| = A_c$ =constant implying that:

$$PEP = \frac{\frac{1}{2}A_c^2}{50} = 2500 \ \ \mathrm{W}$$

Problem 5-27:
Part (a):

- The modulating signal is plotted in the figure below:



- The instantaneous frequency is given by:

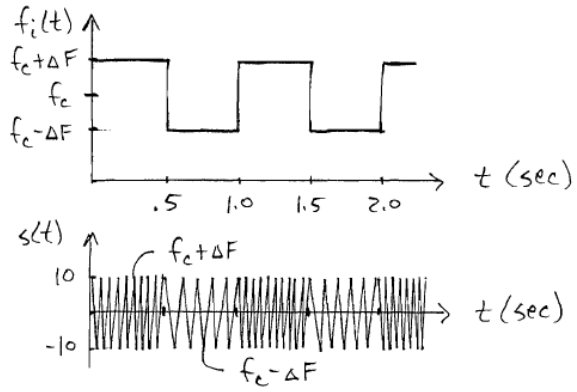
$$f_i(t) = f_c + \frac{1}{2\pi}100m(t)$$

since $m(t)$ is equal to either -5 V or 5 V, then $f_i(t)$ can be written as:

$$f_i(t) = f_c + \begin{cases} \Delta F, & m(t) = 5; \\ -\Delta F, & m(t) = -5. \end{cases}$$

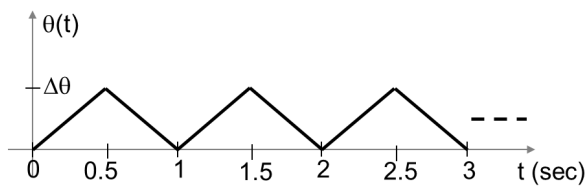
where: $\Delta F = 100 \times 5/(2\pi) = 79.58$ Hz.

- The instantaneous frequency and the FM signal are shown in the figure below:



Part (b):

- The phase deviation is given by: $\theta(t) = 100 \int_{-\infty}^{t} m(\sigma)\mathrm{d}\sigma$. Integrating the rectangular waveform $m(t)$ results in the triangular waveform shown in the figure below:



- The value of $\Delta\theta$ can be obtained by the integration over the interval $[0 \;\; 0.5]$:

$$\Delta\theta = 100 \int_{0}^{0.5} 5\mathrm{d}\sigma = 250 \;\; \text{(rad)}$$

Part (c):

- The peak frequency deviation is given by:

$$\Delta F = \frac{1}{2\pi} D_f \max[m(t)] = \frac{1}{2\pi} \times 100 \times 5 = 79.58 \;\; \text{Hz}$$

note that this value was already calculated in part (a).

Problem 5-28:

- We first use the tone testing to determine the frequency deviation constant D_f:

$$\Delta F = \frac{1}{2\pi} D_f \underbrace{V_{max}}_{=\sqrt{2}V_{RMS}}$$

$$\Rightarrow \; D_f = \frac{2\pi\Delta F}{\sqrt{2}V_{RMS}} = \frac{2\pi \times 30 \times 10^3}{\sqrt{2} \times 1} = 1.333 \times 10^5 \;\; \text{rad/(V.s)}$$

Part (a):

- The frequency modulation index is given by:

$$\beta_f = \frac{\Delta F}{B} = \frac{\frac{1}{2\pi} D_f \max[m(t)]}{f_m} = \frac{(1.333 \times 10^5)(2.5)}{2\pi(1.5 \times 10^4)} = 3.356 \approx 3.5$$

- The amplitude of the FM spectrum at the frequency component $f_c \pm n f_m$ is:

$$S(f_c \pm n f_m) = \frac{A_c}{2} J_n(\beta) = 50 J_n(3.5)$$

  where $f_c = 1$ GHz and $f_m = 15$ kHz.

- The spectral lines that exceed 1% of A_c (i.e. $0.01A_c = 0.01 \times 100 = 1$ V) can be determined from the following table:

n	$J_n(3.5)$	$\lvert S(f_c \pm n f_m)\rvert$
0	-0.380	19.0
1	0.137	6.85
2	0.459	22.95
3	0.399	19.95
4	0.220	11.0
5	0.090	4.5
6	0.029	1.45
7	0.008	0.4

- Consequently, $n \in \{0, 1, 2, 3, 4, 5, 6\}$.

Part (b):

- The frequency modulation index is given by:

$$\beta_f = \frac{\Delta F}{B} = \frac{\frac{1}{2\pi} D_f \max[m(t)]}{f_m} = \frac{(1.333 \times 10^5)(1)}{2\pi(3 \times 10^4)} = 0.7072 \approx 0.7$$

- The amplitude of the FM spectrum at the frequency component $f_c \pm n f_m$ is:

$$S(f_c \pm n f_m) = \frac{A_c}{2} J_n(\beta) = 50 J_n(0.7)$$

  where $f_c = 1$ GHz and $f_m = 30$ kHz.

- The spectral lines that exceed 1% of A_c (i.e. 1 V) can be determined from the following table:

n	$J_n(0.7)$	$\lvert S(f_c \pm n f_m)\rvert$
0	0.881	44.05
1	0.329	16.45
2	0.059	2.95
3	0.007*	0.35

  - Note that the value of $J_3(0.7)$ was determined by the interpolation between 0.6 and 0.8.

- Consequently, $n \in \{0, 1, 2\}$.